

Statement of Work (SOW)

Euphoria Infotech – Sector 5 Kolkata

with

Ion Exchange (India) LTD. – Bangalore

1. Objective

The objective of this engagement is to design, develop, deploy and support an enterprise-grade **AI Monitoring & Optimisation Foundry** that enables real-time monitoring, predictive intelligence, operational optimisation and sustainability management across industrial facilities.

The proposed platform shall provide a unified enterprise solution capable of monitoring electrical systems, water infrastructure, utilities, building services and other intelligent assets through a common digital platform. The solution shall leverage Industrial IoT, Artificial Intelligence, Machine Learning and advanced analytics to provide actionable operational intelligence, predictive recommendations and measurable business outcomes.

The platform shall serve as the foundational digital layer upon which Ion Exchange will deploy its proprietary engineering intelligence, AI models and optimization capabilities, creating a differentiated enterprise offering for industrial customers.

2. Scope of Engagement

Euphoria Infotech shall design, configure, implement and support an enterprise-scale digital platform that provides the foundational capabilities required for the AI Monitoring & Optimisation Foundry.

The platform shall include, but not be limited to:

- Enterprise IoT connectivity
- Industrial data acquisition and normalisation
- Asset onboarding and configuration
- Time-series data management
- Enterprise visualization and dashboards
- Alarm and event management
- Workflow and notification engine
- Reporting and analytics framework
- API management and system integration

- User administration and role-based access control
- Mobile accessibility
- Cloud, On-Premise and Hybrid deployment options

Ion Exchange shall deploy its proprietary engineering templates, AI models and optimization capabilities over this digital foundation.

3. Enterprise Monitoring Foundry

The platform shall function as an enterprise-wide monitoring and intelligence platform capable of supporting multiple industries and diverse asset classes through reusable engineering templates.

The solution shall be capable of monitoring, analysing and optimizing assets including, but not limited to:

Electrical Infrastructure

- HT & LT Panels
- Transformers
- DG Sets
- UPS Systems
- Solar PV Systems
- Battery Energy Storage Systems
- PCC Panels
- APFC Panels
- Smart Energy Meters

Mechanical & Utility Systems

- Pumps
- Compressors
- Motors
- Chillers
- Cooling Towers
- HVAC Systems
- Air Handling Units
- Boilers
- Utility Networks

Water Infrastructure

- STP
- ETP
- RO Plants

- UF Systems
- Softeners
- DM Plants
- Cooling Water Systems
- Chemical Dosing Systems
- Potable Water Networks

Facility & Smart Building Systems

- Lighting Controls
- Fire Detection Systems
- Fire Protection Systems
- Access Control
- Occupancy Monitoring
- Parking Systems
- Environmental Monitoring
- Indoor Air Quality
- Building Automation Systems

The architecture shall support rapid onboarding of additional asset classes without requiring significant application redevelopment.

4. Native AI & Engineering Intelligence

The platform shall seamlessly support integration of Ion Exchange's proprietary Industrial AI capabilities.

These capabilities shall include:

4.1 Asset Templates

Reusable engineering templates containing predefined:

- KPIs
- Engineering parameters
- Alarm philosophies
- Dashboards
- Asset relationships
- Health models
- Maintenance rules
- Optimization logic

The objective is to significantly reduce deployment timelines and standardize implementations across enterprise customers.

4.2 Multi-Variate Anomaly Detection

The platform shall support advanced anomaly detection models capable of analysing multiple operational variables simultaneously, enabling early identification of abnormal equipment behaviour before conventional alarm thresholds are breached.

The capability shall extend across electrical, mechanical, utility and water treatment assets.

4.3 Asset Health Scoring

The solution shall continuously compute an Asset Health Score by evaluating operational conditions, historical performance, maintenance records, environmental factors and equipment utilisation.

Health scores shall be available at asset, plant and enterprise levels.

4.4 Predictive Forecasting

The platform shall provide forecasting capabilities including:

- Energy demand
- Water consumption
- Chemical consumption
- Asset degradation
- Remaining Useful Life (RUL)
- Maintenance schedules
- Utility demand
- Operational performance trends

Forecasts shall support both short-term operational planning and long-term asset management.

4.5 Optimisation Advisories

The platform shall generate engineering recommendations aimed at improving operational efficiency and reducing resource consumption.

Examples include:

- Pump sequencing optimisation
- Chiller performance optimisation
- Cooling tower efficiency improvement
- Boiler combustion optimisation
- HVAC optimisation
- Chemical dosing optimisation
- Energy demand management
- Water recovery enhancement

Wherever possible, recommendations shall quantify expected operational, financial and sustainability benefits.

5. Intelligent Alarm & Event Management

The platform shall provide an intelligent alarm management framework that extends beyond conventional threshold-based notifications.

Each alarm or event shall include, wherever applicable:

- Event description
- Severity classification
- Probable root cause
- Impact assessment
- Affected assets
- Recommended corrective actions
- Estimated energy impact
- Estimated water impact
- Estimated production impact
- Priority level
- Escalation workflow
- Estimated resolution time
- Skill requirements

The objective is to enable maintenance personnel to take informed corrective action with minimal diagnostic effort.

6. Maintenance Workflow Integration

The solution shall support digital maintenance workflows including:

- Alarm acknowledgement
- Work assignment
- Escalation management
- Maintenance checklists
- Mobile work execution
- Photographic evidence
- Root cause documentation
- Closure approval
- Complete audit trail

The platform shall also support integration with enterprise CMMS/EAM solutions where required.

7. Sustainability & Resource Intelligence

The platform shall continuously measure and report enterprise sustainability metrics including:

- Energy savings
- Water savings
- Chemical savings
- Carbon emission reduction
- Wastewater recovery
- Asset efficiency improvement
- Downtime reduction
- Operational cost optimisation

Dashboards shall provide daily, monthly, annual and enterprise-wide performance reporting together with benchmarking across sites a

8. Industrial IoT & Smart Instrumentation

The Vendor shall support integration and deployment of Industrial IoT infrastructure including, where required:

- Smart energy meters
- Water meters
- Flow meters
- Pressure transmitters
- Temperature sensors
- Level sensors
- Vibration sensors
- Power quality analysers
- Water quality analysers
- Wireless sensing devices
- Industrial gateways
- Edge computing devices
- Smart controllers
- Chemical dosing equipment
- Additional instrumentation necessary to achieve agreed project objectives.

9. Enterprise Analytics & Executive Dashboards

The solution shall provide configurable dashboards for multiple stakeholder groups including:

- Executive Management
- Plant Operations
- Maintenance Teams
- Sustainability Teams
- Utility Managers
- Facility Managers

Analytics shall support enterprise benchmarking, KPI tracking, asset performance comparison, trend analysis and operational insights.

10. Integration Framework

The platform shall support secure integration with existing enterprise systems including:

- PLCs
- SCADA
- BMS
- EMS
- DCS
- ERP
- CMMS
- Historians
- Data Lakes
- Third-party applications

The platform shall support open industrial communication standards including OPC-UA, MQTT, Modbus, BACnet, IEC 60870, REST APIs and other commonly adopted enterprise integration protocols.

11. Deployment Architecture

The proposed solution shall support:

- Cloud deployment
- On-Premise deployment
- Hybrid deployment
- Multi-site architecture
- Multi-tenant architecture
- High availability
- Disaster recovery
- Edge computing
- Secure remote access

- Offline data buffering and synchronization

12. Security & Governance

The platform shall incorporate enterprise-grade cybersecurity controls including:

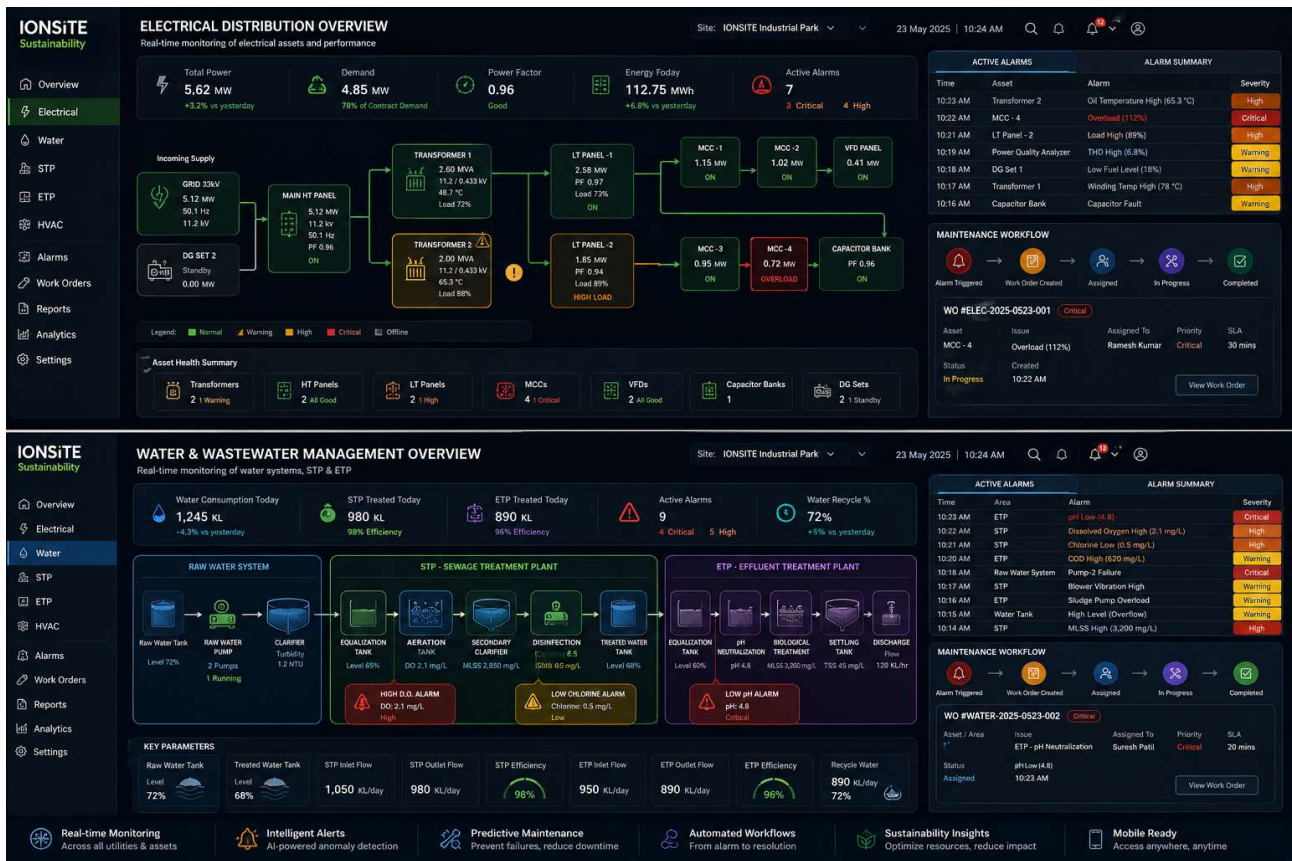
- Role-Based Access Control (RBAC)
- Single Sign-On (SSO)
- Multi-Factor Authentication (MFA)
- Encryption of data at rest and in transit
- Audit logging
- Secure API management
- Backup and recovery
- Compliance with applicable cybersecurity standards

13. Deliverables

The Vendor shall provide all necessary software, platform configuration, integration services, implementation support, documentation, training and post-deployment support required to establish the Enterprise AI Monitoring & Optimization Foundry.

14. The proposed solution is envisioned as an Enterprise AI monitoring and optimisation foundry rather than a conventional BMS system or IOT platform. Euphoria Infotech will provide the underlying digital connectivity, data management and visualisation framework, while Ion Exchange will overlay their engineering domain expertise, AI Models and installations where necessary to create a differentiable Sustainability product.

Reference Dashboard for automatic Alarm Generation



Automated Dashboard for Health Check



End of Document